

Aperture — Low-Level Design

Module: Users

Version: v1.0

Module Owner: User Platform

Dependencies:

- Authentication Module
 - Metadata Module
 - Reviews Module
 - Lists Module
 - Discovery Module
 - Notification Module
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1. Overview

The Users module owns every aspect of a user's identity **after authentication**.

While the Authentication module proves *who the user is*, the Users module defines *who the user becomes* within Aperture.

This distinction is intentional.

Authentication manages identity verification.

Users manages platform identity.

2. Responsibilities

The Users module owns:

- User profile
- Avatar
- Display name
- Biography
- Username
- Favorite movies

- Favorite TV shows
- Favorite directors
- Privacy settings
- Notification preferences
- Following
- Followers
- User statistics
- Public profile pages
- Cinematic profile metadata

The module does **not** own:

- Authentication
 - Passwords
 - Tokens
 - Reviews
 - Lists
 - Recommendations
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3. Functional Requirements

The module shall support:

Profile Management

Users can:

- update profile
 - change username
 - upload avatar
 - edit biography
 - configure preferences
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Public Profiles

Every user has a public profile containing:

- profile information
- recent activity
- favorite films
- favorite shows
- reviews
- lists
- followers

- following
 - viewing statistics
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Follow System

Users can:

- follow
- unfollow
- view followers
- view following

Follow relationships should support future notification and feed generation.

Privacy

Users can control visibility of:

- profile
- reviews
- lists
- activity
- followers

Privacy should be granular rather than global.

4. Module Boundaries

Inputs

- authenticated user
- profile update request
- follow request
- preference updates

Outputs

- public profile
 - user statistics
 - follow relationships
 - profile metadata
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5. High-Level Workflows

Profile Update



Validate Request



Authorize User



Update Profile



Invalidate Relevant Cache



Return Updated Profile

Follow User



Validate Target



Prevent Self Follow



Create Relationship



Publish Activity



Update Counts

↓

Return Success

Profile Request

↓

Resolve Username

↓

Load Profile

↓

Load Statistics

↓

Load Public Activity

↓

Return Profile

6. Public APIs

The Users module provides APIs for:

Profile

- Get profile
- Update profile
- Upload avatar
- Delete avatar

Relationships

- Follow user
- Unfollow user

- Get followers
- Get following

Preferences

- Notification settings
- Privacy settings
- Display preferences

Statistics

- User summary
- Activity overview
- Viewing statistics

Exact endpoints are defined in the API Specification document.

7. Data Model

Primary entities:

User

Stores:

- username
 - display name
 - biography
 - avatar
 - timestamps
 - profile visibility
 - preferences
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Follow Relationship

Stores:

- follower
 - followed user
 - creation timestamp
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User Preferences

Examples:

- language
 - theme
 - spoiler preferences
 - notification preferences
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Statistics are derived rather than permanently stored where practical.

8. Business Rules

Username

- Unique
 - Immutable for a configurable cooldown period
 - Human-readable
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Following

Users cannot:

- follow themselves
 - create duplicate relationships
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Avatars

Initially:

- image upload
- size validation
- format validation

Future:

- automatic optimization
 - cropping
 - multiple resolutions
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9. Validation Rules

Profile

- biography length
- username format
- avatar type
- display name limits

Relationships

- valid target user
- authenticated requester
- relationship uniqueness

Preferences

- valid option values
 - backwards compatibility
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10. Database Interactions

Reads

- profile lookup
- username lookup
- follower counts
- user statistics

Writes

- profile updates
- follow creation
- follow removal
- preference updates

Frequently accessed profile data should be indexed by username.

11. Caching Strategy

Candidate cache entries:

- public profiles

- follower counts
- profile summaries
- user statistics

Cache invalidation occurs after:

- profile edits
- follow changes
- avatar updates

Avoid caching mutable private settings.

12. Performance Considerations

Potential hotspots:

- celebrity profiles with many followers
- popular reviewers
- profile activity feeds

Mitigations include:

- pagination
- caching
- indexed lookups
- asynchronous statistic recalculation

Follower counts should not require scanning relationship tables on every request.

13. Security Considerations

Profile updates require ownership verification.

Private data must never appear in public APIs.

Uploads should validate:

- MIME type
- file size
- supported formats

Future considerations:

- abuse reporting

- impersonation detection
 - username reservation
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14. Integration Points

The Users module integrates with:

Authentication

- authenticated identity
- account lifecycle

Reviews

- author information
- review history

Lists

- ownership
- collaboration

Discovery

- personalization inputs

Notifications

- follow events
- profile activity

Future Recommendation Engine

- Cinematic DNA
 - taste profile
 - viewing preferences
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15. Testing Strategy

Unit Tests

- username validation
- follow rules
- privacy rules

Integration Tests

- profile updates
- follower relationships
- preference persistence

API Tests

- public profile
- authenticated updates
- follow endpoints

Edge Cases

- deleted users
- renamed users
- private profiles
- blocked relationships (future)

16. Future Evolution

Potential additions:

- profile verification
- badges
- achievements
- custom profile themes
- profile banners
- social links
- profile pinning
- public API keys
- account export
- account deletion workflow

The design should allow these features without restructuring the module.

17. Interview Discussion Points

Be prepared to discuss:

- Why separate Authentication and Users?
- Why model follow relationships independently?
- Should follower counts be computed or cached?
- How would you scale profiles with millions of followers?

- How would you implement profile privacy efficiently?
 - What information belongs in the User table versus derived statistics?
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18. Summary

The Users module defines a person's identity within Aperture beyond authentication.

It is intentionally responsible for social identity, preferences, and profile management while remaining independent of authentication and business domains such as Reviews and Lists.

This separation creates a stable foundation upon which the social, discovery, and AI systems can build without introducing unnecessary coupling.